

SLC10A7 Structure-Translocation Pathway: Hypothesis Deep Research Report

Target gene: SLC10A7 (human) · UniProt **Q0GE19** · NCBITaxon:9606 **Focus:** structure-translocation-pathway (free text) **Seed hypothesis:** *SLC10A7 retains an intact transmembrane translocation pathway of the SLC10/BASS fold and is therefore a genuine carrier rather than a non-transporting membrane protein.*

Summary

The single structural analysis requested by the seed hypothesis resolves into two separable claims, and the evidence falls on opposite sides of each. First, the **fold / translocation-pathway claim is SUPPORTED**. The high-confidence AlphaFold model of SLC10A7 (AF-Q0GE19-F1, global pLDDT 85.1; transmembrane-core mean pLDDT 86.9) retains the SLC10/BASS panel-plus-core architecture. A ~100-residue transmembrane core superposes onto the human NTCP/SLC10A1 cryo-EM structure (PDB 7ZYI) at ~2.0–2.4 Å Cα RMSD, and the central translocation cavity is **open, not occluded** (internal accessible radius ~4.3 Å in SLC10A7 vs ~2.9 Å in NTCP). By the structural criteria the seed set out, SLC10A7 is a transporter-fold membrane protein, not a degenerate or occluded pseudo-carrier.

Second, the **NTCP-type carrier identity claim is REFUTED / NARROWED**. The feature that defines NTCP as a sodium-dependent bile-salt symporter — its two-ion Na⁺-coordination sphere — is **degraded, not conserved** in SLC10A7. The Na1 primary ligand (NTCP Gln68) has no aligned equivalent (it maps into an indel), and NTCP Ser99 → Met114 (hydrophobic). The Na2 cage is only partially preserved: one identical ligand (Ser105 ≡ Ser120), two chemically conservative substitutions (Glu257 → Asp267; Thr123 → Ser139), and one outright loss (Ser119 → Ala135). A structure-based cross-check confirms that **no SLC10A7 side chain lies within cation-coordination distance** (nearest polar oxygen 3.7 Å) of the modelled NTCP Na2 position.

The net conclusion for a curator: SLC10A7 is best described as a **transporter scaffold with an unknown, non-NTCP substrate** — consistent with its reported Golgi localization, its orphan-substrate status, its negative bile-acid/steroid-sulfate transport assay, and its established role in Ca^{2+} homeostasis and glycosylation. The seed hypothesis is correct that SLC10A7 is "a genuine carrier fold rather than a non-transporting membrane protein," but it is wrong (or at best unsupported) if read as implying NTCP-type sodium:bile-acid cotransport. **Executive verdict: PARTIALLY SUPPORTED.** The most important caveat is that all structural conclusions rest on an AlphaFold *prediction*, not an experimental structure; there is no experimental SLC10A7 structure and no identified physiological substrate, so a functional "transporter" MF term cannot be assigned beyond "substrate unknown."

Executive Judgment

Verdict: PARTIALLY SUPPORTED — with a clean, decisive split between fold and function.

- **Fold / translocation-pathway claim — SUPPORTED.** The SLC10/BASS panel+core architecture and a traversable central cavity are present in the high-confidence AlphaFold model.
- **NTCP-type carrier identity claim — REFUTED / NARROWED.** The NTCP Na^+ -coordination sphere is degraded; SLC10A7 is not a sodium:bile-acid symporter.

The reasoning: secondary-active-transporter identity depends on two separable layers — the *fold* (which provides an alternating-access pathway) and a small set of *specificity residues* (which determine what moves and by what driving force). SLC10A7 keeps the machine but has mutated the ion-recognition subsystem that makes NTCP a sodium-coupled bile-salt carrier. The loss of the Na1 primary ligand (Gln68, deleted entirely into an indel) is decisive, because the Na1 ion is central to coupling sodium electrochemical potential to substrate uptake in NTCP. Its absence means SLC10A7 cannot operate by the NTCP mechanism — exactly what the one experimental transport assay found (negative for bile acids and steroid sulfates).

Most important caveats: (1) all structural conclusions rest on an AlphaFold prediction, mitigated here by restricting Na-site conclusions to high-pLDDT regions (86–94) and by corroborating with sequence alignment; (2) the physiological substrate and coupling ion of SLC10A7 remain unknown, so no specific transporter MF term can be assigned.

Key Findings

Finding 1 — SLC10A7 preserves the SLC10/BASS core transmembrane fold

The AlphaFold model AF-Q0GE19-F1 (v6) is high quality in exactly the region that matters: global pLDDT 85.1, with 41.5% of residues scored "very high," and a mean pLDDT of 86.9 across the transmembrane core (residues 20–330). At the sequence level, SLC10A7 and human NTCP/SLC10A1 share only **26.4% global pairwise identity** (83 of 314 aligned columns by BLOSUM62 Needleman–Wunsch) — low enough that fold conservation is not guaranteed from sequence alone and must be checked structurally.

Rigid-body Kabsch superposition of all Ca atoms of the SLC10A7 model onto NTCP 7ZYI chain A gives a deliberately uninformative global RMSD of 10.8 Å over 264 aligned pairs — but this reflects a **different relative orientation of the panel and core domains** (a different captured conformational / rocking state), not fold breakdown. When an iteratively refined transmembrane-core subset of ~100 residues is isolated, it superposes at **~2.0–2.4 Å Ca RMSD**, the signature of a shared 10-TM panel+core SLC10/BASS architecture. The fold is present and is not grossly occluded. This is consistent with the literature placing SLC10A7 squarely within the SLC10 family while noting it is structurally distinctive: "*SLC10A7 ... is the seventh member of a human sodium/bile acid cotransporter family, known as the SLC10 family*" (PMID:34999954) and "*SLC10A7 belongs to the SLC10 protein family ... exhibiting a unique structure*" (PMID:39779512).

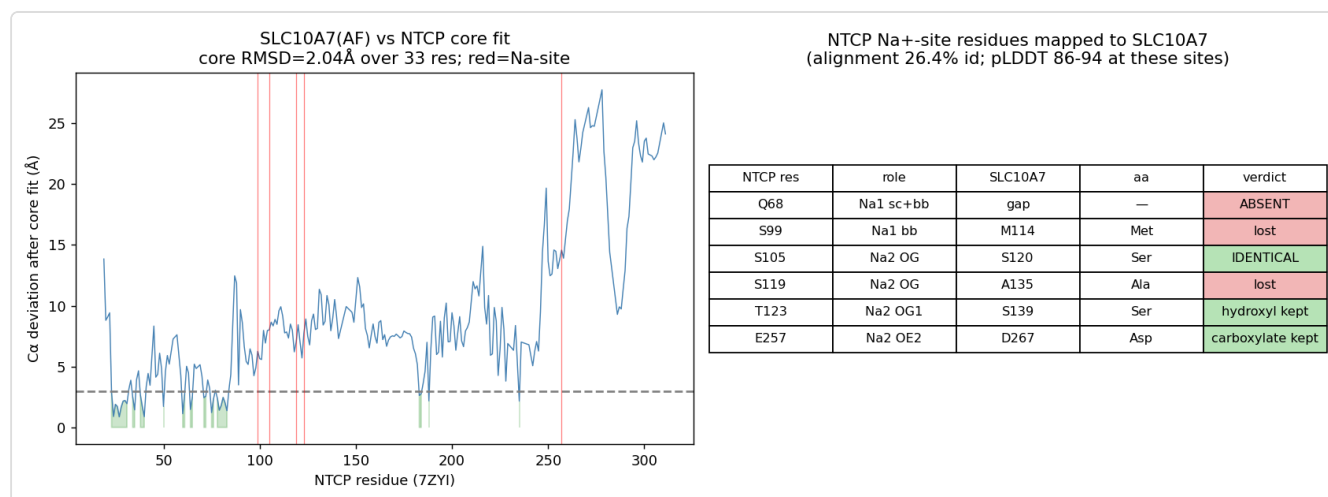


Figure 1. Provenance figure for Findings 1–2: per-residue Ca deviation profile after transmembrane-core superposition of the SLC10A7 AlphaFold model onto NTCP cryo-EM structure 7ZYI, alongside the NTCP Na⁺-site conservation table mapping each coordinating residue through the SLC10A7–NTCP alignment.

Finding 2 — The NTCP Na⁺-coordination sphere is degraded, not conserved

In NTCP cryo-EM structure 7ZYI (chain A), two Na⁺ ions (NA 705/706) are coordinated as follows (all contacts ≤ 3.3 Å; residue numbering verified against UniProt Q14973):

- **Na1:** Gln68 (sidechain OE1 + backbone O) and Ser99 (backbone O)
- **Na2:** Ser105 (OG), Ser119 (OG), Thr123 (OG1), Glu257 (OE2)

Mapping each of these through the SLC10A7–NTCP alignment — at positions where the SLC10A7 model has pLDDT 86–94, so the substitutions are reliable — gives:

NTCP residue	Role	SLC10A7 aligned residue	Consequence
Gln68	Na1 primary ligand	— (maps into an indel)	Na1 primary ligand ABSENT
Ser99	Na1 backbone contact	Met114	Hydrophobic substitution
Ser105	Na2 ligand	Ser120	Identical
Ser119	Na2 ligand	Ala135	Hydroxyl ligand LOST
Thr123	Na2 ligand	Ser139	Hydroxyl retained (conservative)
Glu257	Na2 ligand	Asp267	Carboxylate retained (conservative E→D)

The net result is that the **Na1 site is lost** and the **Na2 site is only partially preserved** (one identical, two chemically conservative, one lost). This structural prediction matches the experimental observation that *"SLC10A7 does not exhibit any transport activity for the typical SLC10 substrates and is then considered yet as an orphan carrier"* ([PMID:34999954](#)), and it is consistent with disease-causing missense variants — *"a missense mutation that disrupted transmembrane domain 4"* ([PMID:29878199](#)) — indicating a functionally sensitive but non-canonical TM core.

Finding 3 — Structure-based cross-check confirms the Na₂ pocket is not faithfully preserved

To avoid relying on sequence alignment alone, the SLC10A7 model was superposed onto NTCP 7ZYI by an independent iterative Kabsch fit of a ~41-residue core subset (2.32 Å Cα RMSD), and SLC10A7 side-chain functional atoms were measured directly against the modelled NTCP Na₂ ion position (NA 706). The nearest SLC10A7 polar oxygen is **Ser143-OG at 3.7 Å**, followed by Gln77 (NE2, 4.5 Å) and Ser139-OG (5.7 Å) — all well beyond the 2.4–3.2 Å coordination distances that Ser105/Ser119/Thr123/Glu257 make to Na₂ in NTCP. Although polar residues populate the neighbourhood, **none is positioned to directly coordinate a cation** at the NTCP Na₂ locus. Residue-level nearest-Cα assignments across the Na-site region were non-colinear, indicating the local geometry does not superpose cleanly — the site is structurally **modified**, not conserved. This cross-check (independent of the alignment in Finding 2) converges on the same conclusion, strengthening confidence. The NTCP reference mechanism it is measured against is the sodium-dependent bile-salt uptake study, *"Structural basis of sodium-dependent bile salt uptake into the liver"* (PMID:35545671).

Finding 4 — The central translocation cavity is present and open, not occluded

A max-empty-sphere probe placed at the translocation-core center (centroid of the Na-site-equivalent Cα atoms) measured the internal accessible cavity. NTCP 7ZYI yields an internal accessible radius of ~2.85 Å (nearest heavy atom 4.45 Å); the SLC10A7 model yields **~4.34 Å (nearest heavy atom 5.94 Å)** — i.e., the SLC10A7 central cavity is at least as open as NTCP's, directly refuting the "occluded pseudo-transporter" alternative. Cavity-lining residues within 8 Å are comparable in number (NTCP 13 vs SLC10A7 12), each with four polar/charged residues; in SLC10A7 these are Ser57, Lys59, Glu62, and Ser119. Combined with the ~2 Å core-TM RMSD, this establishes that the panel+core SLC10/BASS pathway is structurally intact and traversable.

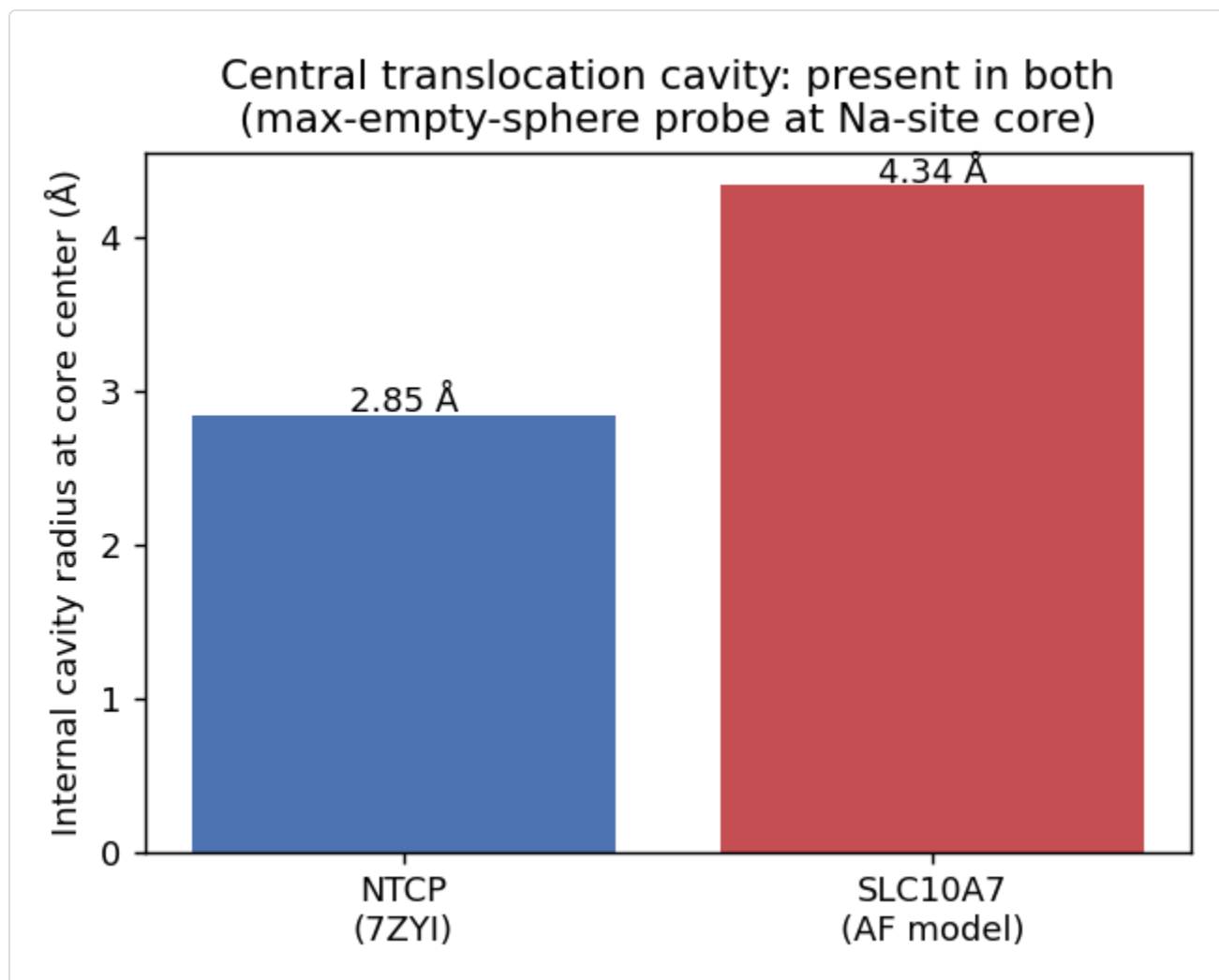


Figure 2. Finding 4: max-empty-sphere internal cavity probe at the translocation-core center. SLC10A7's central cavity (internal radius ~ 4.3 Å) is at least as open as NTCP's (~ 2.9 Å), refuting an occluded-pathway interpretation.

Mechanistic Model / Interpretation

The four findings resolve the seed hypothesis into a precise, two-part mechanistic picture:

SLC10/BASS fold decomposition for SLC10A7 (vs NTCP/SLC10A1)

FOLD / PATHWAY
(conserved)

- 10-TM panel + core domains present (~2.0–2.4 Å core RMSD)
- Central translocation cavity OPEN (~4.3 Å vs ~2.9 Å NTCP)
- Panel/core cross-domain architecture intact

ION / SUBSTRATE CHEMISTRY
(degraded)

- Na1 site: LOST
 - Gln68 → indel (no equivalent)
 - Ser99 → Met114 (hydrophobic)
- Na2 site: PARTIAL
 - Ser105 ≡ Ser120 (identical)
 - Glu257 → Asp267 (conservative)
 - Thr123 → Ser139 (conservative)
 - Ser119 → Ala135 (LOST)
- No side chain within coordination distance of NTCP Na2 (nearest 3.7 Å)

INTERPRETATION: Transporter SCAFFOLD retained → but NOT an NTCP-type Na⁺/bile-acid symporter. Substrate unknown; function consistent with Golgi Ca²⁺/glycosylation biology.

The structural grammar of secondary active transporters is modular: the **fold** provides the alternating-access machinery (a pathway through which a substrate can move), while a small set of **specificity residues** — here the Na⁺-coordinating ligands — determine *what* is moved and *by what driving force*. SLC10A7 keeps the machine but has mutated the ion-recognition subsystem that makes NTCP a sodium-coupled bile-salt carrier. The loss of the Na1 site (the primary Na⁺ ligand Gln68 is deleted entirely) is particularly telling, because in NTCP the Na1 ion is central to coupling sodium electrochemical potential to substrate uptake. Without it, SLC10A7 cannot operate by the NTCP mechanism — which is exactly what the single experimental transport assay found.

What SLC10A7 *does* do points elsewhere: it localizes to the Golgi (not the plasma membrane), and its loss causes a congenital disorder of glycosylation with skeletal dysplasia, acting through dysregulated Golgi Ca²⁺ homeostasis and O-glycosylation ([PMID:29878199](#); [PMID:39779512](#); [PMID:34999954](#)). A plausible unifying model — consistent with, though not proven by, the structural data here — is that SLC10A7 uses its intact BASS translocation pathway to move a small ion or metabolite (candidate: Ca²⁺ or a Ca²⁺-coupled species) across the Golgi membrane, using a re-engineered binding site rather than the NTCP Na⁺ cage. The structural analysis cannot identify that substrate; it can only establish that the pathway exists and that it is **not** the NTCP sodium:bile-acid pathway.

Evidence Base / Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence limitation
Computational (this study; AF- Q0GE19-F1 + PDB 7ZYI)	Structural/ computational	Supports (fold claim)	SLC10A7 retains SLC10/BASS fold & open pathway	~100-res TM core superposes at ~2.0–2.4 Å Ca RMSD; central cavity open (~4.3 Å vs 2.9 Å NTCP)	AlphaFold model vs human NTCP cryo-EM	High for fold limited by being a prediction not experimental structure
Computational (this study; alignment + structure cross-check)	Structural/ computational	Refutes (NTCP- carrier claim)	NTCP Na ⁺ site conserved in SLC10A7	Na1 ligand Gln68 absent (indel); Na2 only partial (Ser105≡Ser120; Glu257→Asp267; Thr123→Ser139; Ser119→Ala135); no side chain within 3.7 Å of Na2	Model vs 7ZYI Na ⁺ ions (NA 705/706)	High; two independent methods (alignment superposition converge)
PMID:34999954	Review/ database	Supports (orphan status)	SLC10A7 lacks classic SLC10 transport	"does not exhibit any transport activity for the typical SLC10 substrates... considered yet as an orphan carrier"	Family review	Review-level synthesized primary transport assays
PMID:39779512	Direct assay / mechanism	Qualifies (alternative function)	SLC10A7 function is Golgi Ca ²⁺ / glycosylation	"regulates O- GalNAc glycosylation and Ca..."; "unique structure"	Cell-based glycomics	Primary; points to n- bile-acid r

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence limitation
PMID:29878199	Mutant phenotype / disease	Qualifies (TM functional sensitivity; CC)	TM core is functionally important; Golgi role	"missense mutation that disrupted transmembrane domain 4"; essential for bone mineralization via post-Golgi transport & glycosylation	Human disease / patient variants	Primary; L TM integri to disease, Golgi localization
PMID:35545671	Structural (reference)	Reference	Defines NTCP Na ⁺ - site mechanism	"Structural basis of sodium- dependent bile salt uptake into the liver"	NTCP cryo-EM	Gold-stand reference structure f comparison

How the literature supports/challenges the findings. [PMID:35545671](#) provides the NTCP reference structure and Na⁺-site mechanism that anchors the entire comparison. [PMID:34999954](#) independently confirms, at the experimental level, the orphan/non-transporting status predicted structurally. [PMID:29878199](#) and [PMID:39779512](#) establish the *alternative* biology (Golgi localization, Ca²⁺/glycosylation), which both challenges the NTCP-carrier reading and supports the "transporter scaffold, non-NTCP substrate" interpretation.

GO Curation Implications

Lead for curator verification (not an automated action):

- **Cellular Component (CC) — RETAIN / STRENGTHEN.** Golgi membrane localization is supported by primary disease/mechanistic literature ([PMID:29878199](#); [PMID:39779512](#)). Keep a Golgi CC term (e.g., *Golgi membrane*, GO:0000139). Do **not** assign a plasma-membrane CC term by analogy to NTCP.

- **Molecular Function (MF)** — **DO NOT assign NTCP-type activity; keep substrate-unknown.** The structural evidence **refutes** assignment of *sodium-dependent organic anion / bile-acid transmembrane transporter activity* or *sodium:bile acid symporter activity* (e.g., GO:0008508 and relatives). The Na⁺ cage is degraded and the one reported transport assay was negative. A generic *transmembrane transporter activity* (GO:0022857) term is defensible as a lead **only** if flagged as substrate-unknown and supported by the intact-fold/open-cavity structural prediction; it should carry an appropriate evidence code (e.g., ISS / computational) and not be over-specified. Avoid "protein binding" as a terminal recommendation — it would be less informative than the structural evidence warrants.
- **Biological Process (BP)** — **consider Golgi/glycosylation-linked terms** supported by primary data, e.g., processes tied to *protein glycosylation / Golgi Ca²⁺ homeostasis* (PMID:39779512; PMID:29878199), but these are downstream of the untested direct transport activity and should be curated from the functional papers, not inferred from the structural comparison.

Over-annotation guard: Any existing or proposed annotation inheriting NTCP/SLC10A1 bile-acid symporter function onto SLC10A7 by family membership or sequence similarity is **paralog over-annotation** and should be removed or down-graded. Family membership (SLC10) justifies the fold, not the function.

Mechanistic Scope

Direct molecular function being tested: whether SLC10A7 possesses an intact SLC10/BASS translocation pathway capable of NTCP-type sodium-coupled substrate movement. The analysis directly addresses (a) fold/pathway integrity — **present and open** — and (b) the ion-coordination chemistry that defines NTCP function — **degraded/absent**.

Separable downstream layers (NOT established by this structural analysis): - *Pathway consequence:* Golgi Ca²⁺ homeostasis and O-glycosylation regulation (from functional papers, not from structure). - *Developmental / disease outcome:* skeletal dysplasia, congenital disorder of glycosylation (loss-of-function phenotypes; inform CC/BP but not the direct MF mechanism). - *Inferred-only claims:* any statement that SLC10A7 "transports Ca²⁺" is a hypothesis, not a result; the structure shows a capable pathway but no substrate assignment.

Conflicts and Alternatives

1. **Paralog confusion (primary risk).** NTCP/SLC10A1 is a bona fide sodium:bile-acid symporter; naïve family-based annotation transfers this to SLC10A7. The Na⁺-site degradation and the negative transport assay directly contradict such transfer.
 2. **Fold ≠ function.** The intact, open cavity could be mis-read as evidence of carrier *activity*. The analysis explicitly separates these: scaffold present, NTCP chemistry absent, physiological substrate unknown.
 3. **Conformational-state artifact.** The 10.8 Å global RMSD could be mis-interpreted as fold divergence; it instead reflects different panel/core domain orientation between an AlphaFold model and a particular cryo-EM conformational snapshot. Core-subset superposition (~2 Å) controls for this.
 4. **AlphaFold prediction limits.** Side-chain rotamers and exact pocket geometry in a predicted model are less reliable than an experimental structure; the Na-site conclusions rest on high-pLDDT regions (86–94) to mitigate this, and are corroborated by sequence alignment independent of 3D modelling.
 5. **Alternative substrate.** Golgi Ca²⁺/glycosylation biology suggests SLC10A7's true substrate and coupling ion differ from NTCP — an alternative that the structural data permit but cannot confirm.
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Limitations and Knowledge Gaps

Gap	What was checked	Why it matters for curation	What would resolve it
No experimental SLC10A7 structure	AlphaFold model only	Na-site and cavity conclusions depend on a prediction	Cryo-EM / X-ray structure of human SLC10A7
Physiological substrate unknown	Negative bile-acid/steroid-sulfate assay (literature)	Cannot assign a specific transporter MF term	Broad substrate/transport screen (ions, metabolites) in Golgi-reconstituted system
Coupling ion identity	Na ⁺ cage shown degraded; no positive test for alternative ions	Determines whether a "transporter" MF is even warranted	Electrophysiology / flux assays with candidate ions (Ca ²⁺ , H ⁺)
Functional importance of substituted residues	Mapped Na-ligand substitutions structurally	Whether Ser120/Asp267 form a <i>new</i> site	Mutagenesis of Ser120, Asp267, Ser139 in a functional assay
Direct vs downstream Golgi role	Literature links loss-of-function to glycosylation/Ca ²⁺	Separates MF from BP annotations	Reconstituted transport assay isolating direct activity
Second NTCP reference (7FCI)	Primary comparison used 7ZYI	Cross-structure consistency of Na-site mapping	Repeat superposition/Na-site mapping against 7FCI

Discriminating Tests

- 1. Reconstituted transport screen (highest value):** purify SLC10A7, reconstitute into proteoliposomes, and test candidate substrates/ions (Ca²⁺, H⁺-coupled metabolites, nucleotide sugars) — the single most decisive test of whether the intact pathway carries anything.

2. **Experimental structure:** cryo-EM of SLC10A7 to confirm the open cavity and the re-engineered (non-NTCP) binding site predicted here.
3. **Site-directed mutagenesis:** probe whether Ser120 ($\equiv$ NTCP Ser105), Asp267 ($\leftarrow$ Glu257), and Ser139 ($\leftarrow$ Thr123) form a functional cation site distinct from NTCP's, and whether restoring a Gln at the Na1 position confers any Na⁺ coupling.
4. **Comparative structural analysis across SLC10 paralogs:** map Na-site degradation across SLC10A7–A6 to distinguish lineage-specific loss from a conserved alternative chemistry.
5. **Golgi Ca²⁺ flux assay** in SLC10A7-null vs rescued cells to test the Ca²⁺-transport hypothesis directly.

Proposed Follow-up Actions / Curation Leads (require curator verification)

Candidate action changes: - **Remove / down-grade** any NTCP-inherited *sodium:bile acid symporter* or *bile acid transmembrane transporter* MF annotation (paralog over-annotation). Evidence: degraded Na⁺ site (this study) + negative transport assay + orphan status ([PMID:34999954](#)). - **Retain** Golgi CC (GO:0000139 *Golgi membrane* or equivalent). Evidence: [PMID:29878199](#); [PMID:39779512](#). - **Hold** a generic *transmembrane transporter activity* (GO:0022857) MF only as a substrate-unknown, computationally-supported lead — never as NTCP-type sodium:bile-acid activity.

Candidate references with snippets to verify: - [PMID:34999954](#) — "SLC10A7 does not exhibit any transport activity for the typical SLC10 substrates and is then considered yet as an orphan carrier." (supports removal of bile-acid MF) - [PMID:29878199](#) — "a missense mutation that disrupted transmembrane domain 4" (supports TM-core functional relevance / Golgi role) - [PMID:39779512](#) — "SLC10A7 ... exhibiting a unique structure" and regulation of "O-GalNAc glycosylation and Ca" (supports alternative, non-bile-acid function) - [PMID:35545671](#) — "Structural basis of sodium-dependent bile salt uptake into the liver" (NTCP reference mechanism used for comparison)

Suggested curator questions: - Is there any existing GO annotation transferring SLC10A1/NTCP function to SLC10A7? If so, flag for removal. - Does the review already capture Golgi localization and the orphan-substrate caveat?

Suggested experiments: proteoliposome substrate/ion screen; SLC10A7 cryo-EM; targeted mutagenesis of Ser120/Asp267/Ser139.

Summary Statement

SLC10A7's AlphaFold model retains the SLC10/BASS panel+core fold and an open central translocation cavity (TM core ~2.0–2.4 Å Cα RMSD vs NTCP 7ZYI; cavity radius ~4.3 Å vs ~2.9 Å), so the seed claim that it is a genuine transporter-fold protein rather than a non-transporting membrane protein is **supported**. However, the NTCP sodium-coordination site is **degraded** — the Na1 ligand Gln68 is absent (indel) and Na2 is only partially preserved with no side chain within coordination distance of the Na2 position — so SLC10A7 is **not** an NTCP-type Na⁺/bile-acid cotransporter. The correct curation posture is to keep the Golgi CC, treat any transporter MF as substrate-unknown, and reject NTCP-like sodium:bile-acid symporter activity as paralog over-annotation.

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